

Tutorial - Inferential Statistics

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Problem 1

Two random samples are drawn from two normal populations.
Their values are:

$A : 65, 66, 73, 80, 82, 84, 88, 90, 92$

$B : 64, 66, 74, 78, 82, 85, 87, 92, 93, 95, 97$

Test whether the two populations have the same variance at the 5% level of significance.



Problem 2

A county environmental agency suspects that the fish in a particular polluted lake have elevated mercury level. To confirm that suspicion, six striped bass in that lake were caught and their tissues were tested for mercury. For the purpose of comparison, five striped bass in an unpolluted lake were also caught and tested. The fish tissue mercury levels in mg/kg are given below.

Sample 1 (from polluted lake)	Sample 2 (from unpolluted lake)
0.610	0.429
0.703	0.391
0.582	0.570
0.591	0.573
0.621	0.497
0.551	

- (a) Construct the 95% confidence interval for the difference in the population means based on these data.
- (b) Test, at the 5% level of significance, whether the data provide sufficient evidence to conclude that fish in the polluted lake have elevated levels of mercury in their tissue.



Problem 3

We intend to compare several methods for predicting the shear strength for steel items. Data for two of these methods, the M1 and M2, when applied to nine specific items, are shown in Table 1.

Determine if there is a significant difference in shear strength predictions between methods M1 and M2 at a significance level of $\alpha = 0.05$. Address the following while arriving at the conclusion:

- (a) Formulate the Null and Alternate Hypothesis
- (b) Calculate the test statistic and compare with the critical value
- (c) Also compute the p value corresponding to the test statistic

Item	Method 1	Method 2
1	1.186	1.061
2	1.151	0.992
3	1.322	1.063
4	1.339	1.062
5	1.200	1.065
6	1.402	1.178
7	1.365	1.037
8	1.537	1.086
9	1.559	1.052

Table 1: Comparison of predictions of Method 1 and Method 2 for 9 items



Problem 4

A new diagnostic test is designed to detect a rare disease. A test result of positive indicates the presence of the disease, while a negative result suggests its absence. The hypotheses are as follows:

H_0 : The patient does not have the disease.

H_1 : The patient has the disease.

- (a) Define Type-I and Type-II errors in this context and explain the potential impacts of each error on the patient.
- (b) Given the rarity of the disease, why might the designers of the test prioritize minimizing the Type-I error? What implications does this have for the probability of Type-II error?
- (c) Suppose the probability of a Type-I error (false positive) is 0.01. If the probability of a Type-II error (false negative) is 0.2, what is the test's power?
- (d) If the prevalence of the disease is very low (e.g., 0.1%), discuss how Type-I and Type-II errors might impact the test's overall reliability. How would you interpret a positive result?

